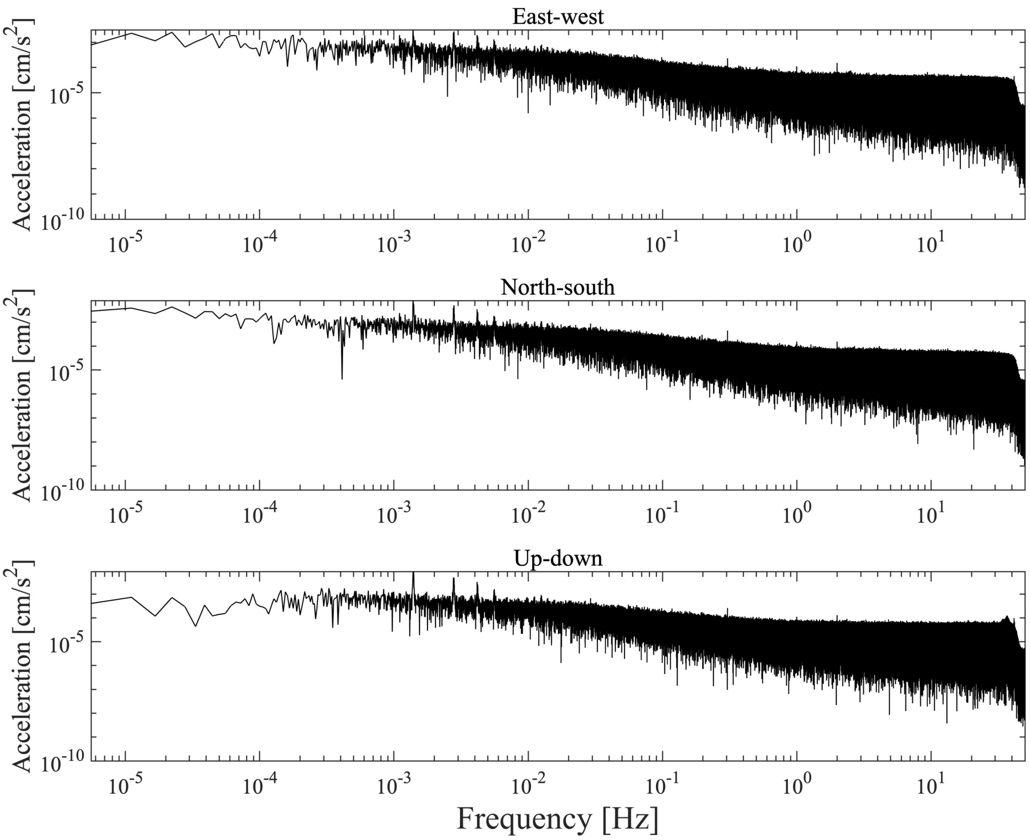
Supplementary Material

1 Supplementary Data

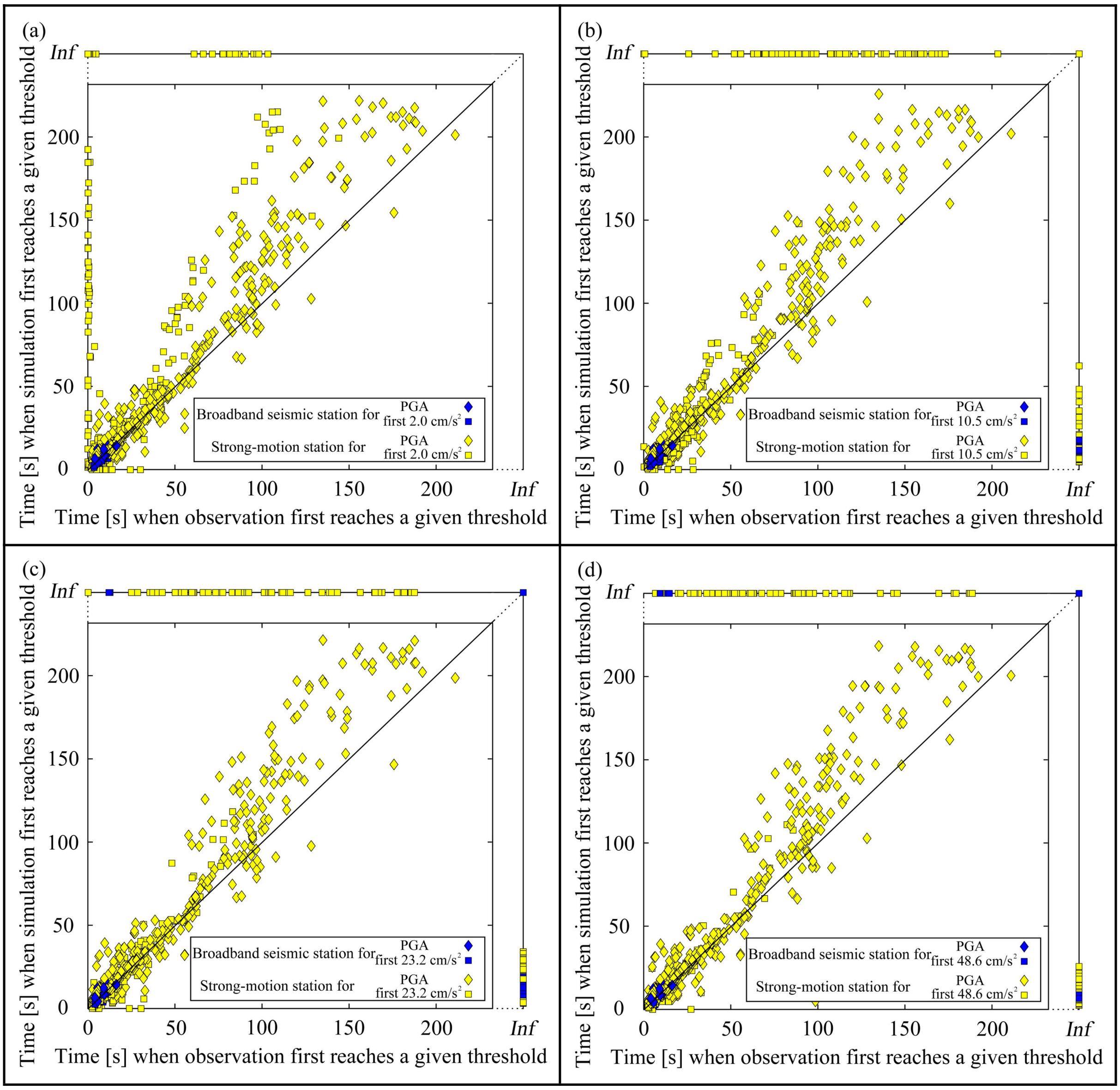
Please see Google link [https://drive.google.com/drive/folders/1wMJxZUdWJLgqCrMNHpBXUUQ73z53C0HF?usp=sharing] or Baidu link [https://pan.baidu.com/s/1_6WpGlI4L-wWYGZpH1oH-w; Passwords: 92mt] for the three-component waveforms/envelopes of observations and simulations of 58 earthquakes in Table 1. Please see Goole link [https://drive.google.com/drive/folders/1TSpggWNYPIcsIO8JfidOPMySEfzXkoy_?usp=sharing] and Baidu link [https://pan.baidu.com/s/1hlqzmTL8mBkvT5rTZ3GX4w; Passwords: wiw8] for the outputs of FinDer for dataset 1 and dataset 2 of the 58 earthquakes. The file name of the folders is the year, month, day, hour, minute and second of the earthquake occured. In the folder for each case, the sub folder named info contains the parameters of the output results, and the ones named temp contains the visualization plots based on info.

2 Supplementary Figures and Tables

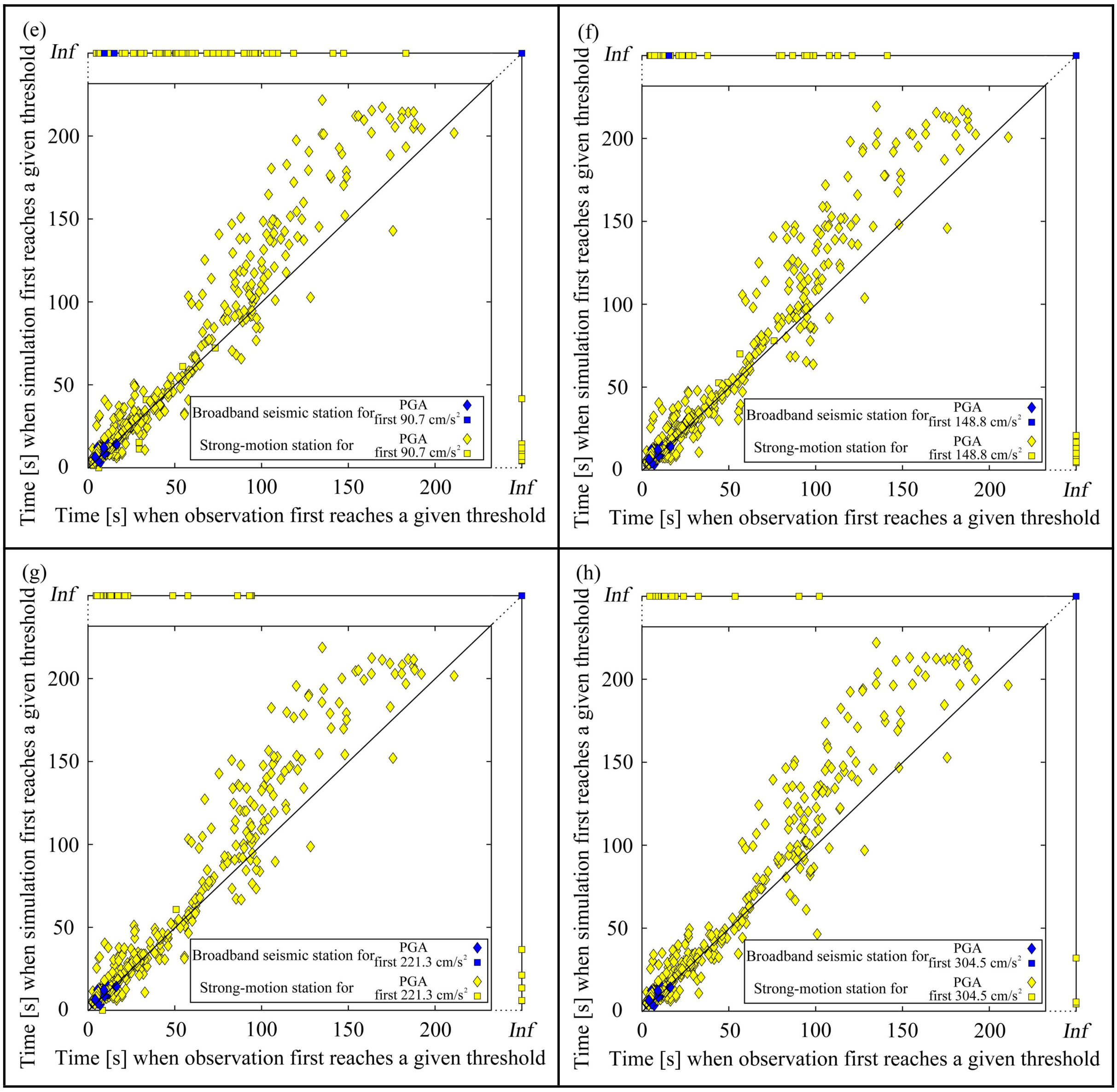
2.1 Supplementary Figures



Supplementary Figure 1. The three-component traces in frequency domain of noise continuously observed by a low-cost intensity sensor deployed at 23.67°N and 116.64°E with around 50 hours of noise records starting from February 19, 2019.



Supplementary Figure 2. Comparison of the 58 studied earthquakes in terms of the time when observations and simulations first reach the given thresholds. These values correspond to FinDer's trigger and template cropping thresholds. The *Inf* means that the three-component records of the station do never reach the given thresholds. Follows Figure 4.



Supplementary Figure 2. Continued.

2.2 Supplementary Tables

Supplementary Table 1. Reference line-source parameters for 6 events with M_W greater than 6.0 (marked grey in Table 1). Each line-source model is extracted from the distribution of aftershocks of 1 day after the mainshock. The rupture length, $L_{\_a}$, is the length from one end-point to the other. Magnitude M_{W_a} is calculated from $L_{\_a}$ according to the Equation (1).

yyyy-mm-dd hh:mm:ss	M_{W_a}	$L_{\_a}$ [km]	End-point1		End-point2		middle-point	
			lat[°]	lon[°]	lat[°]	lon[°]	lat[°]	lon[°]
2008-05-12 14:28:04	8.0	255	30.89	103.19	32.52	105.08	31.71	104.13
2013-04-20 08:02:47	6.8	39	30.04	102.83	30.35	103.02	30.20	102.93
2014-08-03 16:30:12	6.7	34	26.95	103.45	27.18	103.22	27.07	103.34
2014-10-07 21:49:39	6.9	53	23.23	100.34	23.57	100.70	23.40	100.52
2014-11-22 16:55:28	6.3	20	30.18	101.74	30.33	101.63	30.26	101.69
2017-08-08 21:19:48	6.4	25	33.31	103.74	33.09	103.81	33.20	103.78

Supplementary Table 2. The velocity crust model of Sichuan-Yunnan region, China, from Wang et al. (2003).

Depth [km]	V_P [km/s^2]	V_S [km/s^2]	V_P/ V_S
0-2	4.11	2.4	1.71
2-5	5.31	3.06	1.74
5-10	5.97	3.44	1.74
10-20	6.25	3.61	1.73
20-42	6.40	3.70	1.73

Supplementary Table 3. Testing FinDer trigger parameters for the 2008 Wenchuan M_S 8.0 earthquake.

Trigger threshold [cm/s^2]	Minimum number of trigger stations: 1		2		3			4	
2	3	14	12	13	13	20	80	19	103
	249	213	249	213	213	249	213	249	213
	7.9	7.8	7.9	7.8	7.8	7.9	7.8	7.9	7.8
	45	45	45	45	50	45	50	45	40
4.6	30		12		12			12	
	249		249		249			249	
	7.9		7.9		7.9			7.9	
	50		55		55			55	
10.5	18		12		12			12	44
	213		213		213			213	291
	7.8		7.8		7.8			7.8	8
	65		65		65			65	55
23.2	45		45		45			45	
	291		291		291			291	
	8		8		8			8	
	55		55		55			55	

The four numbers from top to bottom in each grid give the FinDer trigger time [s], rupture length [km], magnitude and strike. Multiple columns of data in a single grid represent split events.

Supplementary Table 4. Source parameters (columns 2 & 3) and FinDer line-source models (L, M_{FD}, Θ) for all events in dataset 1 and 2. The number in the first column corresponds to Table 1. D_{epi} gives the average distance from the epicenter to the three closest stations.

No.	M_S/M_W	Strike1/dip1/rake1 Strike2/dip2/rake2 [º]	Dataset 1				Dataset 2			
			D_{epi} [km]	L [km]	M_{FD}	Θ [º]	D_{epi} [km]	L [km]	M_{FD}	Θ [º]
1	8.0/7.9	231/35/138 357/68/63	35	249	7.9	55	8	213.2	7.8	45
2	5.5/5.3	212/29/54 71/67/108	27	15.4	6.1	30	15	15.4	6.1	100
3	5.0/4.9	206/41/83 34/49/96	28	--	--	--	13	5.2	5.4	145
4	6.3/5.7	204/84/-3 294/87/-174	33	6.1	5.5	5	11	28.6	6.5	110
5	5.4/5.2	21/83/4 291/86/173	40	--	--	--	12	13.2	6.0	130
6	5.2/4.9	350/40/82 180/50/96	65	--	--	--	13	9.7	5.8	110
7	5.0/4.9	286/84/180 16/90/6	23	--	--	--	13	7.1	5.6	45
8	5.0/4.9	43/44/80 236/47/99	30	--	--	--	7	18	6.2	65
9	5.2/5.2	14/76/-9 106/81/-166	47	--	--	--	7	9.7	5.8	35
10	5.0/5.1	236/79/176 327/86/11	45	--	--	--	10	7.1	5.6	30
11	5.0/5.0	14/44/59 233/53/116	36	--	--	--	10	4.5	5.5	75
12	5.4/5.4	313/75/-7 45/83/-165	17	28.6	6.5	15	5	13.2	6.0	160
13	5.3/5.0	271/44/82 103/46/98	32	--	--	--	12	13.2	6.0	120
14	5.2/5.1	251/86/1 161/89/176	17	11.3	5.9	130	6	6.1	5.5	120
15	5.2/5.0	232/44/70 79/50/108	16	28.6	6.5	110	7	9.7	5.8	115
16	5.7/5.6	313/46/-126 179/55/-59	25	9.7	5.8	160	11	15.4	6.1	60
17	5.7/5.6	350/51/35 236/63/136	18	9.7	5.8	125	9	15.4	6.1	80
18	5.6/5.3	234/58/155 338/69/34	17	24.5	6.4	30	6	11.3	5.9	80
19	5.5/5.6	208/86/179 298/89/4	112	--	--	--	57	--	--	--
20	5.5/5.4	337/42/-113 187/52/-71	29	4.5	5.4	120	11	15.4	6.1	65

21	5.1/5.3	332/52/-131 206/54/-50	10	9.7	5.8	20	10	11.3	5.9	120
22	7.0/6.6	212/42/100 19/49/81	23	98.5	7.3	175	6	45.4	6.8	175
23	5.0/-	-/-/-/ -/-/-	19	13.2	6.0	175	9	6.1	5.5	35
24	5.4/5.4	215/45/100 21/46/80	19	9.7	5.8	15	7	13.2	6.0	95
25	5.0/4.8	177/42/74 17/50/103	23	18	6.2	20	10	6.1	5.5	20
26	5.4/5.2	35/45/86 221/45/94	19	13.2	6.0	60	8	9.7	5.8	80
27	5.2/5.2	66/50/-125 294/51/-55	77	--	--	--	6	33.4	6.6	30
28	5.9/5.7	97/42/-95 284/48/-85	30	45.5	6.8	5	8	24.5	6.4	130
29	5.1/4.9	360/45/45 235/60/125	19	--	--	--	11	15.4	6.1	165
30	6.6/6.2	71/81/-175 340/86/-9	21	13.2	6.0	175	8	45.5	6.8	135
31	5.2/5.1	317/78/5 226/86/168	18	13.2	6.0	95	14	9.7	5.8	160
32	5.2/5.2	254/66/169 349/80/25	39	--	--	--	13	9.7	5.8	160
33	6.9/6.1	329/81/174 60/84/9	18	15.4 53.1	6.1 6.9	165 165	16	33.4	6.6	5
34	6.4/6.1	143/85/-1 233/89/-175	38	18	6.2	40	21	15.4	6.1	85
35	5.9/5.7	238/89/179 328/89/1	20	39	6.7	115	14	24.5	6.4	130
36	5.9/5.6	79/72/9 346/81/162	25	21	6.3	170	17	11.3	5.9	165
37	5.9/5.5	339/71/173 71/84/19	24	45.5	6.8	175	18	11.3	5.9	10
38	5.0/4.9	158/44/89 340/46/91	28	--	--	--	11	11.3	5.9	70
39	5.5/5.3	69/66/9 336/82/156	76	--	--	--	14	11.3	5.9	80
40	5.0/4.9	177/45/-118 35/51/-64	29	--	--	--	10	5.2	5.4	15
41	5.1/5.0	293/86/177 23/87/4	28	--	--	--	14	6.1	5.5	125
42	5.2/5.2	281/47/-55 56/53/-122	54	--	--	--	41	--	--	--
43	5.2/5.2	288/47/-61 69/51/-117	55	--	--	--	42	--	--	--
44	5.1/5.1	318/85/179 48/89/5	42	--	--	--	10	9.7	5.8	145
45	7.0/6.5	151/79/-8	38	18	6.2	175	24	33.4	6.6	170

		243/82/-168								
46	5.4/5.1	21/54/42 263/57/135	27	24.5	6.4	20	9	11.3	5.9	160
47	5.1/5.1	201/61/3 110/88/151	12	9.7	5.8	20	7	8.3	5.7	20
48	5.0/4.9	206/77/9 113/81/167	12	8.3	5.7	95	7	6.1	5.5	35
49	5.9/5.7	126/80/-178 36/88/-10	72	--	--	--	20	24.5	6.4	95
50	5.3/5.1	75/83/-176 344/86/-7	38	--	--	--	11	7.1	5.6	65
51	5.1/5.0	183/84/2 92/88/174	22	6.1	5.5	75	12	8.3	5.7	135
52	5.7/5.3	79/81/-174 348/84/-9	55	--	--	--	8	11.3	5.9	65
53	5.1/5.0	349/41/43 223/63/122	34	--	--	--	10	13.2	6.0	140
54	6.0/5.7	184/40/123 323/57/65	33	53.1	6.9	150	8	21	6.3	80
55	5.1/5.1	178/43/115 326/52/69	22	53.1	6.9	95	9	6.1	5.5	175
56	5.3/4.8	155/33/116 305/61/74	32	--	--	--	8	28.6	6.5	85
57	5.4/5.2	343/45/71 190/48/108	18	24.5	6.4	170	9	13.2	6.0	25
58	5.6/5.5	11/39/75 210/52/102	21	28.6	6.5	135	9	18	6.2	140